

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

MEERA – ONLINE JEWELLERY E-COMMERCE PLATFORM

1. PROJECT OBJECTIVES

The primary objective of the **Meera Online Jewellery E-Commerce Platform** is to provide a modern, secure, and user-friendly digital platform for browsing, discovering, and purchasing jewellery products online. The system aims to simplify the overall shopping process while providing customers with useful personalized and support features.

The key objectives of the system are:

- To provide a centralized digital platform for showcasing and selling jewellery products through a well-structured catalogue.
- To organize jewellery into categories and provide search, filtering, and detailed product information for efficient product discovery.
- To provide a simple and seamless shopping experience from product selection to checkout and order placement.
- To implement secure user registration, authentication, login, and account management.
- To provide essential e-commerce features including **shopping cart, wishlist, checkout, and order management**.
- To securely maintain customer profiles, delivery addresses, wishlist information, and order records in a structured database.
- To provide personalized shopping experiences through **Style Finder** and **AI-based Custom Hampers**.
- To provide accessible customer support through FAQs, shipping and return information, size guides, and jewellery care instructions.
- To ensure a responsive and user-friendly interface across different devices and screen sizes.
- To maintain **data security, authentication, database-level access control, scalability, and reliability** for future system enhancements.

2. KEY FEATURES

The Meera platform incorporates the following key features:

- **Product Catalogue**

A structured jewellery catalogue allows users to explore products across multiple categories, including necklaces, earrings, rings, bracelets and bangles, bridal sets, and kids' collections.

- **Product Discovery**

Users can browse products, access product details, and use search and filtering mechanisms to identify jewellery according to their preferences.

- **User Authentication**

The platform provides secure user registration, login, logout, and session management through an integrated authentication system.

- **Shopping Cart**

Users can add products to their cart, modify quantities, remove products, and review the total value before proceeding with checkout.

- **Wishlist**

Users can save products to a personal wishlist for future reference or purchase.

- **Checkout and Order Management**

The system provides a structured checkout process through which users can provide delivery information and place orders. Users can subsequently access their order information and status.

- **User Profile and Address Management**

Authenticated users can manage their profile information and maintain shipping or billing addresses associated with their account.

- **Personalized Shopping**

The platform provides interactive features such as **Style Finder** and **AI Custom Hampers**, enabling users to explore personalized jewellery and gifting options.

- **Customer Support**

Dedicated support resources provide information related to frequently asked questions, shipping, returns, jewellery sizing, and jewellery care.

- **Company Information**

The platform includes informational sections such as About, Careers, and Press, providing users with relevant organizational and brand information.

- **Responsive User Interface**

The application follows a responsive, mobile-first design approach, allowing users to access the platform across desktops, tablets, and mobile devices.

- **Administrative Capabilities**

Authorized administrative functionality supports the management and monitoring of application-related data and customer orders.

- **Data Security**

Database operations are protected through authentication and **Row Level Security (RLS)** policies to restrict unauthorized access to user-specific information.

3. FUNCTIONAL REQUIREMENTS

Functional requirements define the specific operations and services that the Meera system shall provide to its users.

FR-01: User Registration

- The system shall allow new users to create an account using the registration interface.
- The system shall validate the information submitted during registration.
- The system shall securely create and maintain the user's authentication record.
- The system shall provide appropriate feedback when registration is unsuccessful.

FR-02: User Authentication

- The system shall allow registered users to log in using their credentials.
- The system shall authenticate users before providing access to protected account features.
- The system shall maintain the user's authenticated session.
- The system shall provide a logout mechanism.
- The system shall prevent unauthorized users from accessing protected resources.

FR-03: Product Catalogue Management

- The system shall display available jewellery products.
- The system shall organize products into predefined categories.
- The system shall provide individual product detail views.
- The system shall display relevant product information such as product name, images, price, and description.
- The system shall allow users to browse products efficiently.

FR-04: Product Search and Filtering

- The system shall allow users to search for relevant products.
- The system shall provide filtering or category-based navigation where applicable.
- The system shall display products according to the selected criteria.

FR-05: Shopping Cart Management

- The system shall allow users to add products to the shopping cart.
- The system shall allow users to modify product quantities.
- The system shall allow users to remove products from the cart.
- The system shall calculate the applicable cart total.
- The system shall display the selected products before checkout.

FR-06: Wishlist Management

- The system shall allow authenticated users to add products to their wishlist.
- The system shall allow users to remove products from their wishlist.
- The system shall maintain wishlist information for the respective user.
- The system shall provide access to saved products for future reference.

FR-07: Checkout and Order Processing

- The system shall provide a checkout interface for placing orders.
- The system shall collect the required customer and delivery information.
- The system shall validate the information required for order placement.
- The system shall create and store order information after successful order submission.
- The system shall associate orders with the respective authenticated customer.

FR-08: Profile and Address Management

- The system shall allow authenticated users to access their profile.
- The system shall allow users to update applicable profile information.
- The system shall allow users to add, update, and manage delivery addresses.
- The system shall associate address information with the corresponding user account.

FR-09: Personalized Features

- The system shall provide a Style Finder feature for assisting users in discovering suitable jewellery.
- The system shall provide an AI Custom Hamper feature for creating personalized gifting options.
- The system shall store applicable custom hamper information for authenticated users.

FR-10: Customer Support

- The system shall provide an FAQ section for commonly asked questions.
- The system shall provide shipping-related information.
- The system shall provide return and refund-related information.
- The system shall provide jewellery size guidance.
- The system shall provide jewellery care instructions.
- The system shall provide a mechanism for submitting customer enquiries or contact requests.

FR-11: Administrative Functions

- The system shall provide administrative access to authorized personnel.
- The administrator shall be able to manage relevant product and order information.
- The administrator shall be able to monitor customer orders.
- Administrative functionality shall be restricted to authorized users.

FR-12: Data Management and Security

- The system shall store application data in the configured database.
- The system shall maintain user, address, wishlist, contact, custom hamper, and order information.
- The system shall apply appropriate access-control policies to database resources.
- The system shall restrict users from accessing data belonging to other users.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes, operational constraints, and performance expectations of the Meera system.

NFR-01: Performance

- The system should provide responsive interaction and navigation across the application.
- Product information and application pages should load within an acceptable time under normal operating conditions.
- Database operations should be performed efficiently to provide a smooth user experience.

NFR-02: Usability

- The application shall provide a clear, intuitive, and user-friendly interface.
- Navigation and product discovery should require minimal user effort.
- Important functions such as cart, wishlist, checkout, login, and profile management should be easily accessible.
- Error and validation messages should be clear and understandable.

NFR-03: Security

- The system shall authenticate users before granting access to protected functionality.
- User-specific information shall be protected from unauthorized access.
- Database access shall be controlled through appropriate security policies.
- Row Level Security (RLS) shall be used to protect applicable database records.
- Administrative functionality shall be restricted to authorized personnel.

NFR-04: Reliability

- The system should perform consistently under normal operating conditions.
- User and transaction-related information should be stored accurately.
- The system should handle invalid input and application errors without compromising stored data.
- Database operations should maintain data consistency.

NFR-05: Availability

- The application should remain accessible whenever the hosting environment and dependent services are operational.
- Core shopping functionality should be available without unnecessary service interruptions.

NFR-06: Scalability

- The system architecture should support the addition of new products and jewellery categories.
- The application should be capable of accommodating an increasing number of users and transactions.
- The architecture should permit integration of additional services and e-commerce functionality in future releases.

NFR-07: Maintainability

- The application shall follow a structured and modular architecture.
- Components, pages, authentication logic, and database operations should remain logically separated.
- The system should be easy to modify, debug, and enhance.
- Future developers should be able to understand and maintain the application with minimal difficulty.

NFR-08: Compatibility

- The application should support modern web browsers.
- The user interface should function across commonly used desktop and mobile devices.
- The responsive design should maintain usability across different screen sizes.

NFR-09: Responsiveness

- The interface shall adapt appropriately to desktop, tablet, and mobile screen dimensions.
- Navigation menus and interactive components should remain usable across supported devices.

NFR-10: Data Integrity

- The system shall maintain consistency between users, addresses, wishlist records, custom hampers, and orders.